

Differential Privacy



COMPCSI 752: Big Data Management

University of Auckland

(Credits to Ninh Pham)

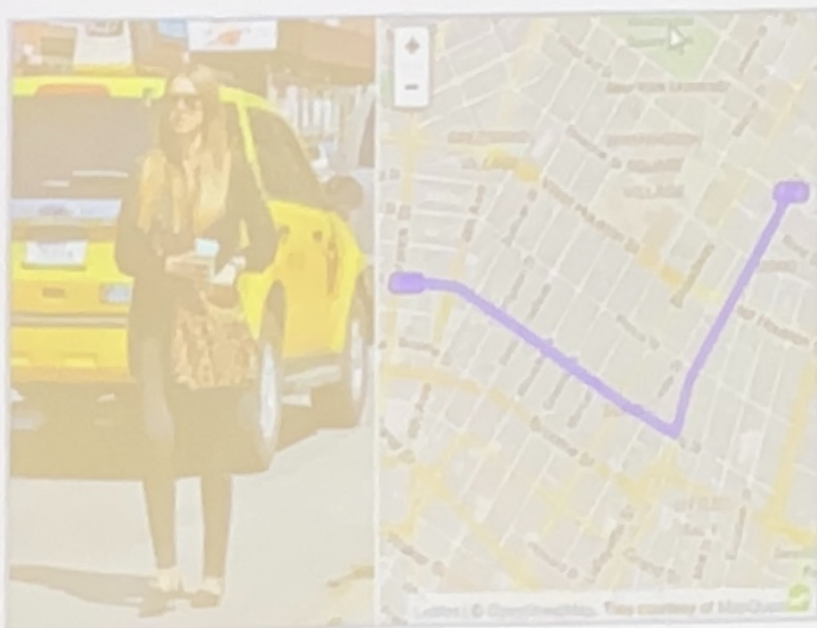
Slides are collected and edited from many resources in the reference

Auckland, May 2025

Outline

- Motivation
- Why is anonymization hard?
- Differential privacy
- Applications

Data release horror stories



Jessica Alba (actor)

- NYC Taxi and Limousine Commission released anonymized data
- De-anonymize:
 - Google images of a star getting out of a taxi (e.g. celebrity gossip blogs)
 - Cross-reference with released data
- Alba took a taxi from Trump SoHo, and did not add a tip to \$9 fare!

Data release horror stories

Matching Known Patients to Health Records in Washington State Data



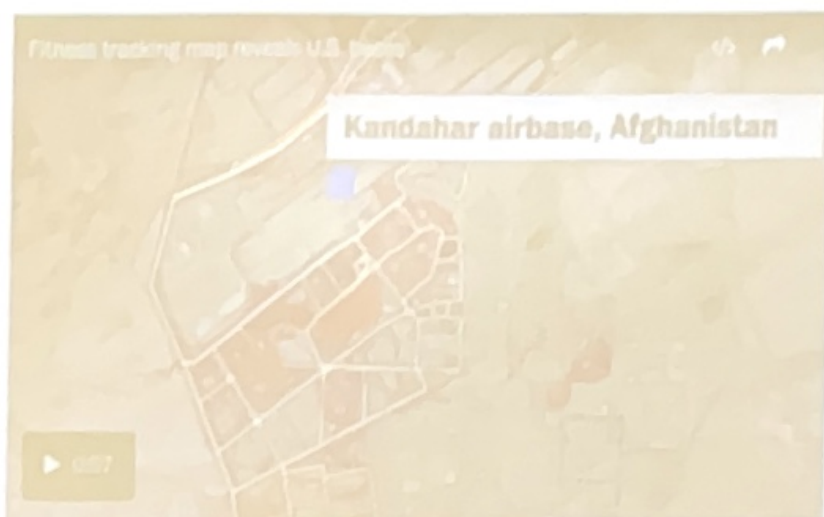
Information from news accident reports uniquely and exactly matched medical records in publicly available Washington State health data in 43% of the cases, thereby putting names to patient records.

Data release horror stories

Strava's fitness tracker heat map reveals the location of military bases

Geolocation isn't a new problem for the military

By [Andrew Lapins](#) | [@AndrewLapins](#) | Jan 28, 2018, 3:51pm EST



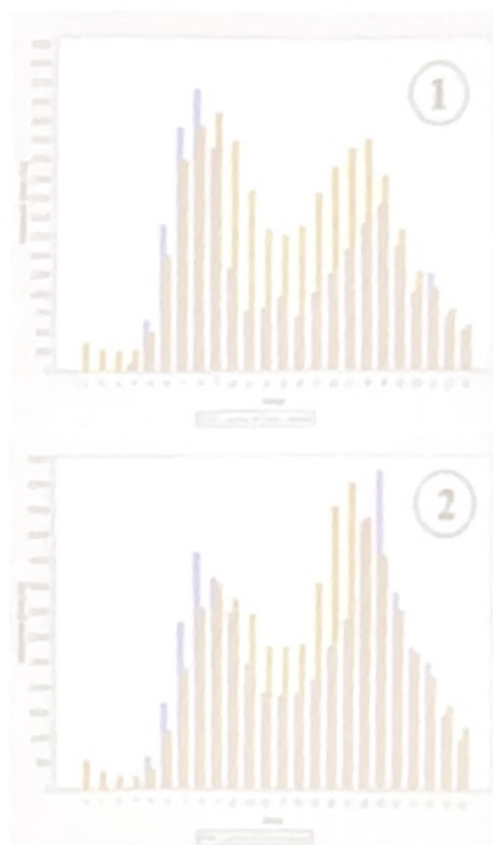
Commonwealth of Massachusetts
Group Insurance Commission

*Your
Benefits
Connection*

87% of all Americans can be uniquely identified using 3 bits of information: ZIP code, birth date, and gender.

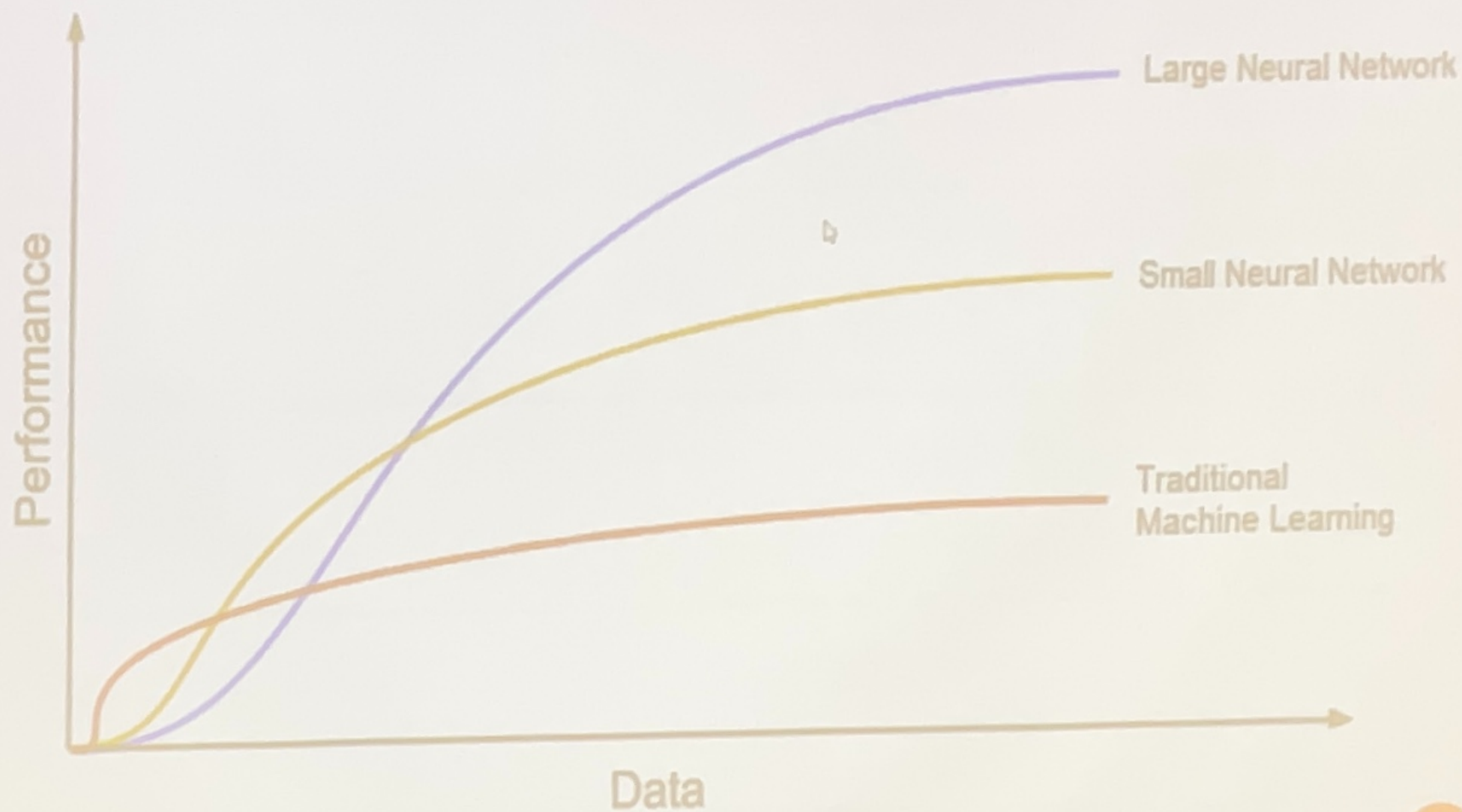
We need to solve this data release problem...

Why release data?



- Urban mobility modelling at scale
 - Smart traffic design is critical since congestion gets worse

Deep learning & big data



Big data to solve big problems

- Computation with big data:
 - Statistical correlation (genetic association)
 - Aggregate statistics (web analytics)
 - Identify events (disease outbreaks)
 - Data mining
- Individual has useful data but wants privacy
- How can we work on sensitive data?
 - Easy answer: Anonymize it and share
 - Problem: How to do that?



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Defence against privacy attacks

- Anonymization techniques:

- Remove personally identifiable information PII
- K-anonymization

与 $k-1$ 个其他个体看起来相同,

- Encryption techniques:

- Homomorphic encryption
- Multi-party computation

同态加密

Anonymization technique

- Principle:

- Remove identifying information in the data before releasing
- E.g. Sensitive columns (i.e. Name and Address) are masked

- Can we compute on anonymized data, i.e. not including personally identifiable information?

not really

User ID	Name	Address	Account	Sub. Date
001	Alice	123 A St	Pro	01/02/20
002	Bob	234 B St	Free	02/03/21
003	Charlie	456 C St	Pro	03/04/18

User ID	Name	Address	Account	Sub. Date
001			Pro	01/02/20
002			Free	02/03/21
003			Pro	03/04/18

Linkage attack

- Netflix challenge:
 - Netflix released its database for the \$1 million contest
 - Sanitization: Personal identities removed
- Sparsity of data:
 - No two Netflix records are similar more than 50% with high probability
 - If profiles can be matched up to 50% to a profile in IMDB, then adversary knows the true identity of the profile w.h.p.

NETFLIX



Personally identifiable information (PII)

- PII definitions (from Wikipedia):
 - US: Information which can be used to distinguish or trace an individual's identity, such as their name, social security number, biometric records, etc. alone, or when **combined** with other personal or identifying information which is linked or linkable to a specific individual, such as date and place of birth, mother's maiden name,...
 - EU: 'personal data' shall mean any information relating to an identified or identifiable natural person ('data subject'); an identifiable person is one who can be identified, directly or indirectly, in particular by **reference** to an identification number or to one or more factors specific to his physical, physiological, mental, economic, cultural or social identity;
- Problem: Without releasing entire data, how can we compute some innocent **statistics** for analysis?

Attacks from published statistics

Learning Your Identity and Disease from Research Papers: Information Leaks in Genome Wide Association Study

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Abstract

Genome-wide association studies (GWAS) aim at discovering the association between genetic variations, particularly single-nucleotide polymorphism (SNP), and common diseases, which have been well recognized to be one of the most important and active areas in biomedical research. Also renowned is the privacy implication of such studies, which has been brought into the limelight by the recent attack proposed by Homer et al. Homer's attack demonstrates that it is possible to identify a participant of a GWAS from analyzing

1. INTRODUCTION

The rapid advancement in genome technology has revolutionized the field of human genetics by enabling the large-scale applications of genome-wide association study (GWAS) [7], a study that aims at discovering the association between human genes and common diseases. To this end, GWAS investigators determined the genotypes of two groups of participants, people with a disease (cases) and similar people without (controls) in an attempt to use statistical testing to identify genetic markers, typically single-nucleotide

Database privacy



- Need a formal privacy definition since ad-hoc solutions do not work
- How can we have privacy-preserving queries?

Blending into a crowd

- “I am safe in a group of K or more!”
 - K could vary in (3, ... 100, ... , 5.1M)



- Reasons:
 - Rare properties help re-identify someone
 - Privacy is “protection of being brought to the attention of others”

The Yale Law Journal

Volume 89, Number 3, January 1980

Privacy and the Limits of Law

Ruth Gavison†

Anyone who studies the law of privacy today may well feel a sense of uneasiness. On one hand, there are popular demands for increased protection of privacy, discussions of new threats to privacy, and an intensified interest in the relationship between privacy and other

Cluster-based definition

- **K-anonymity:**

- Attributes are *suppressed* or *generalized* until each row is identical to at least $K - 1$ other rows. The database is called **K-anonymous**.
- Linkage attack will result in a group of **K** records

- **Methods:**

- **Suppression:** Replace individual attributes with a *
- **Generalization:** Replace individual attributes with a broader category (e.g. office: CBD $\Rightarrow$ office: Auckland)

lose detail

Why is anonymization hard?

- High-dimensional/high-resolution data is essentially unique
 - Significant computations to achieve **K**-anonymity

<i>office</i>	<i>department</i>	<i>date joined</i>	<i>salary</i>	<i>d.o.b.</i>	<i>nationality</i>	<i>gender</i>
London	IT	Apr 2015	£###	May 1985	Portuguese	Female

- Low-dimension/low-resolution is more private, but less useful in analysis

<i>office</i>	<i>department</i>	<i>date joined</i>	<i>salary</i>	<i>d.o.b.</i>	<i>nationality</i>	<i>gender</i>
UK	IT	2015	£###	1980-1985	—	Female

要么 概括 → 隐私.

less useful

第4 详细 $\rightarrow$ 应用.

Encryption techniques

- Encryption:
 - A cryptography approach converts the original representation of information (plaintext) into an alternative form (ciphertext)
- Common techniques:
 - Homomorphic encryption (HE)
 - Secure multi-party computation (MPC)



Homomorphic encryption (HE)

- Allow computations on encrypted data without decrypting it
 - Use a **public key** to encrypt data, and apply algebraic system (e.g. addition, multiplication) to allow computations on ciphertext
 - Only person with **matching private key** can access the decrypted results

